

FITBUDDY

AI Fitness Plan Generator using Gemini Models

SOFTWARE REQUIREMENTS DOCUMENT (SRS)

1. Introduction

1.1 Purpose

This Software Requirements Document describes the functional and non-functional requirements of FitBuddy, an AI-powered web application that generates general personalized fitness plans using Google Gemini models.

1.2 Scope

FitBuddy allows users to enter fitness-related details and goals, then generates a structured fitness plan with workout suggestions. The application is intended for general fitness guidance and does not provide medical diagnosis or replace advice from qualified health professionals.

1.3 Intended Users

The intended users are individuals who want general workout organization and fitness guidance, including beginners and people with different activity levels.

2. Overall Description

2.1 Product Perspective

FitBuddy is a web-based application with a user interface, a server-side component, and integration with the Gemini API. The server processes user requests and communicates with the AI service.

2.2 Product Functions

- Accept user inputs such as age range, fitness goal, activity level, available workout time, and preferences.
- Validate required fields and display helpful messages for missing or invalid inputs.
- Send the relevant request to the Gemini API through the backend.
- Generate and display a structured fitness plan with exercise suggestions, duration, and schedule guidance.
- Show clear error messages when the network or AI service is unavailable.
- Allow users to review or regenerate a plan by changing their inputs.

3. System Requirements

3.1 Functional Requirements

ID	Requirement
FR-01	The system shall provide a form for entering user fitness details.
FR-02	The system shall check that required input fields are completed and valid.
FR-03	The system shall submit requests to the backend for processing.
FR-04	The backend shall securely call the Gemini API using a server-side API key.

ID	Requirement
FR-05	The system shall display the generated fitness plan in a readable format.
FR-06	The system shall inform users when generation fails and provide a retry option.
FR-07	The system shall not expose API credentials in the client-side interface.

3.2 Non-Functional Requirements

Category	Requirement
Usability	The interface should be simple, clear, and easy for beginners to use.
Performance	The application should show a loading indicator while a plan is being generated.
Security	The Gemini API key must be stored in a server-side environment variable and must not be committed to public source code.
Reliability	The system should handle invalid inputs, API errors, and network failures gracefully.
Compatibility	The web interface should work on commonly used modern browsers and adapt to laptop and mobile screens.
Maintainability	The code should be organized into manageable frontend and backend components.

4. Hardware and Software Requirements

4.1 Hardware Requirements

- Computer or laptop with a modern processor
- Minimum 4 GB RAM (8 GB recommended)
- Internet connection
- Keyboard, mouse, and display

4.2 Software Requirements

- Operating System: Windows, Linux, or macOS
- Frontend: HTML, CSS, JavaScript (or the selected frontend framework)
- Backend: Node.js and Express (if used in the implementation)
- AI Service: Google Gemini API
- Browser: Current version of Chrome, Edge, Firefox, or another modern browser
- Development tools: Visual Studio Code and a package manager such as npm

5. Assumptions and Constraints

- Internet access is required to communicate with the Gemini API.
- Gemini API access, valid credentials, quotas, and applicable terms are required.
- AI-generated plans may need user review and should be treated as general information.
- Users with injuries, medical conditions, or special health needs should consult a qualified professional before following a plan.

6. Expected Outcome

The completed system is expected to accept fitness details, request a plan from Gemini through a secure backend, and present the generated suggestions in a clear and user-friendly web interface.

Project Title: FitBuddy – AI Fitness Plan Generator using Gemini Models